

Title: NED Multi-Wavelength Extragalactic Physics: AGN Luminosity Functions and UQFF Buoyancy-Modified Hubble Tension Analysis

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: NED_BASE, NED_API, QUASAR_SDSS)

Index Slot: 1.10 Database Integration & Multi-Wavelength Astrophysics,
\$n = [\text{int}]\$ PAPER #78 Extragalactic Physics: NED Multi-Wavelength + UQFF

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Abstract

The NED (NASA/IPAC Extragalactic Database) multi-wavelength catalog covers UV through radio for >1 billion extragalactic objects. Key physics tests for UQFF: (1) AGN luminosity functions comparing UQFF-enhanced accretion vs standard models, (2) the Hubble tension ($H_0 = 6773 \text{ km/s/Mpc}$) examined through the UQFF Buoyant vacuum correction, and (3) quasar absorption line systems (DLA/LLS) testing the UQFF vacuum density at cosmological redshifts.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.010^{+4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. UQFF Hubble Constant Analysis

Hubble Tension Context

Measurement	H_0 (km/s/Mpc)	Method
Planck 2018 (CMB)	67.4 ± 0.5	Early universe
SH0ES 2023 (Cepheids)	73.0 ± 1.0	Distance ladder
Tension	4.2σ	

UQFF Buoyant Correction to H_0

The UQFF vacuum buoyancy modifies the effective expansion rate:

$$H_{\text{UQFF}}(z) = H_0 \times \sqrt{\Omega_\Lambda + \Omega_m(1+z)^3 + [UA] \times \rho_{\text{vac, (UQFF)}} \times 8\pi G/3H_0^2}$$

The $[UA] = 0.0001$ fractional vacuum coupling adds:

$$\Delta H_0 = H_0 \times [UA] \times 0.5 = 67.4 \times 0.0001 \times 0.5 = 0.0034 \text{ km/s/Mpc}$$

UQFF correction to Hubble tension: $H_0 = 0.003$ km/s/Mpc far too small to resolve the 5.6 km/s/Mpc tension. The UQFF does not attempt to resolve Hubble tension through the basic [UA] coupling; a higher-order Resonant Hubble correction would require additional development.

2. AGN Luminosity Function Comparison

The UQFF Superconductive mode modifies the AGN accretion efficiency, shifting the break luminosity L_* :

$$L_*^{\text{UQFF}} = L_*^{\text{standard}} \times (1 + [SCm]) = L_* \times 1.99$$

NED quasar catalog comparison:

Redshift bin	L_* _standard ($L_?$)	L_* _UQFF ($L_?$)	NED data range
$z = 0.5$	$10^{\{45.0\}}$	$10^{\{45.3\}}$	$10^{\{44.8\}10^{\{45.5\}}$
$z = 1.0$	$10^{\{45.5\}}$	$10^{\{45.8\}}$	$10^{\{45.2\}10^{\{46.0\}}$
$z = 2.0$	$10^{\{45.8\}}$	$10^{\{46.1\}}$	$10^{\{45.5\}10^{\{46.3\}}$
$z = 3.0$	$10^{\{46.0\}}$	$10^{\{46.3\}}$	$10^{\{45.7\}10^{\{46.5\}}$

UQFF L_* shift of 0.3 dex lies within the 0.5-dex observed scatter compatible with NED quasar data at all redshifts.

3. Quasar Absorption Line Systems

DLA (Damped Lyman- α) systems contain high column density neutral hydrogen ($N_{\text{HI}} > 210 \text{ cm}^{-2}$). The UQFF predicts no modification to the HI 21 cm line frequency (only gravitational Doppler at $10^?$ level). NED DLA catalog: UQFF-consistent.

Summary

Observable	NED Data	UQFF Prediction	Agreement
H_0 tension	5.6 km/s/Mpc	$H_0 = 0.003$ (negligible)	Not resolved
AGN L_*	$10^{\{45\}10^{\{46.5\}}$	+0.3 dex ([SCm])	Within scatter
DLA HI column	$N_{\text{HI}} > 10^{\{20\}}$	Unmodified	Compatible

Source: `QCalc_validation.py` NED_BASE endpoint | $? = 0.0005/\text{day}$ | $[SSq] = 0.57$

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete – 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 igl[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr)imesigl(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1+10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.